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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

MANAGEMENT OF THE DIABETIC PARTURIENT FOR ELECTIVE CAESAREAN SECTION

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1. Introduction and Purpose

- Pregnant patients with diabetes mellitus can be classified as follows :

1. Pre-existing diabetes mellitus
   a) Insulin dependent diabetes mellitus (Type 1 diabetes mellitus)
   b) Non-insulin dependent diabetes mellitus (Type 2 diabetes mellitus) – Diet or tablet controlled (Tablet controlled are converted to insulin during pregnancy)

2. Gestational diabetes
   a) Insulin dependent
   b) Non-insulin dependent - diet controlled

- Tight glycaemic control is essential perioperatively to avoid episodes of hypoglycaemia, hyperglycaemic ketoacidosis and wound infection. Regional anaesthesia is safer than general anaesthesia as the patient can usually warn you sooner of any symptoms of hypoglycaemia they may be experiencing.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the management of diabetic patients undergoing elective Caesarean section.

3. Points to consider

- The patient should be assessed by an anaesthetist prior to surgery so that a discussion of the type of anaesthetic to be administered can occur, and a plan for the management of her diabetes can be made.

- The patient should be placed first on the theatre list if possible

- Patients should be fasted for 6 hours but allowed sips of water until theatre (max 30ml/hour for the last 2 hours prior to theatre)

- **Pre-Op carbohydrate drinks should not be administered to <u>insulin dependent</u> diabetic patients**

- Antacid prescription : Ranitidine 150mg at 22:00 (previous day), Ranitidine at 07:00 (on day of surgery for morning list or 12:00 for afternoon list) and 30ml 0.3M Sodium Citrate immediately prior to attending theatre

- The patient should take their normal early morning dose of soluble insulin with breakfast at 07:30 on the day of surgery for an afternoon theatre list. No further subcutaneous insulin should be taken preoperatively. For morning lists no insulin should be given that morning since the patient is already fasting.

- On arrival of the patient in hospital her blood sugar (BM) should be measured

- Patients with Type 1 diabetes mellitus on subcutaneous insulin pumps should discontinue using their pumps on arrival in hospital

- A **Variable Rate Intravenous Insulin Infusion Regime (VRIII) *** should be commenced if the BM > 7.8mmol/l. BM's need to be checked hourly while on the VRIII

- Surgery should not commence until a blood sugar in the range 4 – 7.8 mmol/l is achieved

- During surgery under general anaesthesia the BM should be checked ½ hourly, but hourly is satisfactory under regional anaesthesia

- If the patient is having the procedure performed under regional anaesthesia the BM should be checked hourly, and the patient should be asked to alert the anaesthetist if she has any hypoglycaemic symptoms.

- Following surgery a BM should be checked in recovery.

- Postoperatively a BM should be checked hourly while on the VRIII, or if not on VRIII then initially two hourly, and if stable then four hourly.

- Once the patient is eating and drinking normally the **Variable Rate Intravenous Insulin Infusion Regime*** can be discontinued.

- **Insulin dependent diabetic patients** can then be placed on their routine insulin regime. However only half the pre-delivery dosage should be prescribed initially

- **Gestational diabetic patient** will usually not require any diabetic medication post delivery

- The Diabetes Specialist Nurse should be contacted for advice if :

  a) There is a delay in an IDDM patient being able to eat and drink and hence recommence their normal insulin regime

  b) Any diabetic patient who has poorly controlled post delivery blood glucose levels

- For poorly controlled diabetics seek advice from a Diabetologist

* Variable Rate Intravenous Insulin Infusion Regime

- Commence an intravenous infusion of :

  500ml bag of 0.9% Saline + 5% Glucose + 10mmol KCl (0.15%) at a rate of 50mls/hr

- Commence an insulin infusion (50iu actrapid in 50 ml 0.9% saline) at a variable rate (see below)

	DOSING ALGORITHM (Please see the guide below)			ALGORITHM GUIDE
Algorithm >	**1**	**2**	**3**	• **ALL** women with diabetes should have Capillary Blood Glucose (CBG) testing hourly in established labour and at least once on admission for induction of labour or elective C-section • Start VRIII and Fluids if CBG > target (see below) or at the start of established labour if the woman has type 1 diabetes **Algorithm 1** Most women will start here **Algorithm 2** Use this algorithm for women who are likely to require more insulin (on steroids; on >80 units of insulin during pregnancy; or those not achieving target on algorithm 1) **Algorithm 3** Use this for women who are not achieving target on algorithm 2 (No patient starts here without diabetes or medical review) If the woman is not achieving targets with these algorithms, contact the diabetes team (out of hours: Medical SpR on call)
	For most women	For women not controlled on algorithm 1 or needing >80 units/day of insulin	For women not controlled on algorithm 2 (after specialist advice)	
CBG Levels (mmol/L)	Infusion Rate (units/hr = ml/hr)			
<4	STOP INSULIN FOR 20 MINUTES Treat hypo as per guideline (re-check CBG in 10 minutes)			
4.0 – 5.5	0.2	0.5	1.0	
5.6 – 7.0	0.5	1.0	2.0	
7.1 – 8.5	1.0	1.5	3.0	
8.6 – 11.0	1.5	2.0	4.0	
11.1 – 14.0	2.0	2.5	5.0	
14.1 – 17.0	2.5	3.0	6.0	
17.1 – 20.0	3.0	4.0	7.0	
>20.1	4.0	6.0	8.0	
Signed				**Target CBG level = 4 – 7.8 mmol/L**
Print Name				**Check CBG every hour whilst on VRIII and every half an hour if under anaesthesia**
Date				**Move to the higher algorithm** if the CBG is > target and is not dropping **Move to the lower algorithm** if CBG falls below 4 mmol/L or is dropping too fast

NB. For safety reasons please ensure that the same intravenous cannula (or a CVP line) is used to infuse both the insulin and the glucose. A Y-connector with a one way valve should always be used and attached onto the intravenous cannula. The bag of 0.9% Saline + 5% glucose + 0.15% KCl infusion should always be attached to a one way valve port to avoid retrograde infusion of insulin into the glucose bag.

4. References

1. Management of glycaemic control in pregnant women with diabetes on obstetric wards and delivery units (2017). Joint British Diabetes Societies for Inpatient Care
2. Oxford Handbook of Anaesthesia, First edition (2001) by K.G.Allman & I.H.Wilson